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Summary

Section 1 Systems of Linear Equations

1.1 Forming the Equations

We read the context of the question and form equations depending on the context of the question.

Example 1

$$\left.\begin{aligned} a+b+c&=176 && —(1)\\ 0.85a+0.90b+0.95c&=161.80 && —(2)\\ 0.93a+0.95b+0.98c&=169.56 && —(3)\end{aligned}\right\} \Rightarrow a=32,\ b=44,\ c=100$$

Example 2

The "variables" may be expressions. To find x, y and z, use the given information to form the equations:

$$\left.\begin{aligned} 2x^3+3\ln y-e^z&=18 && —(1)\\ x^3-\ln y+2e^z&=9 && —(2)\\ x^3+2\ln y+e^z&=11 && —(3)\end{aligned}\right\} \Rightarrow A=8,\ B=1,\ C=1$$

It is technically still 'linear' because we can think of $A=x^3$, $B=\ln y$, $C=e^z$, which turns it into a linear system in A, B and C. A truly nonlinear equation could look like $x^2+x+y^2+2y-3z^3-xz+xy=1$; it cannot be made linear this way, because x appears as both x^2 and x, and there are product terms such as xz and xy.

1.2 Special Case: Number of Equations > Number of Variables

Sometimes the number of equations is more than the number of variables, and the solution must satisfy all of the equations.

Example 3

$$\left.\begin{aligned} \lambda+\mu&=1 && —(1)\\ 2\lambda-\mu&=5 && —(2)\end{aligned}\right\} \Rightarrow \lambda=2,\ \mu=-1$$

$$\lambda+3\mu=-1 \qquad —(3)$$

Sometimes there may be no solution, e.g. skew lines in vectors. In this case, solve for λ and μ using equations (1) and (2). Then sub into (3).

> **Note**
>
> Check (3): if LHS = RHS, the equations are consistent; if LHS ≠ RHS, there is no solution.

1.3 Special Case: Number of Equations < Number of Variables

Sometimes the number of equations is less than the number of variables, so there are infinitely many solutions.

$$\begin{aligned}\text{Sub into (3): LHS} &= 2+3(-1)\\ &= -1\\ &= \text{RHS}\\ \therefore \lambda = 2,\ \mu &= -1\end{aligned}$$

Example 4

$$\left.\begin{aligned} x+y+z&=6 \quad —(1)\\ x-y+2z&=5 \quad —(2)\end{aligned}\right\} \Rightarrow x=\tfrac{11}{2}-\tfrac{3}{2}z,\ y=\tfrac{1}{2}+\tfrac{1}{2}z,\ z=z$$

> **Note**
>
> Useful trick: in contextual questions, the variables are usually non-negative integers, $z \in \mathbb{Z}_0^+$, so use this to find the solution from the general solution.

When $z=z$, it means that there are infinitely many solutions, and z can be any real value. So just sub that into x and y to get a correct solution.

$z=-1$: $(7,0,-1)$ $z=0$: $(\tfrac{11}{2},\tfrac{1}{2},0)$ $z=1$: $(4,1,1)$ $z=2$: $(\tfrac{5}{2},\tfrac{3}{2},2)$

$z=3$: $(1,2,3)$ $z=4$: $(-\tfrac{1}{2},\tfrac{5}{2},4)$ $z=5$: $(-2,3,5)$ $z=6$: $(-\tfrac{7}{2},\tfrac{7}{2},6)$

Section 2 Inequalities

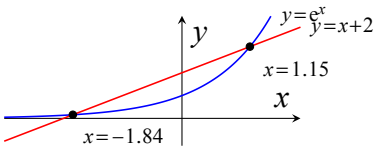
2.1 Solving Inequalities Graphically

Some inequalities cannot be solved algebraically, so we sketch the graphs and read off the solution.

- Rearrange to $f(x) > 0$, or compare $y = f(x)$ with $y = g(x)$.
- Sketch on the GC; mark the intersections and the asymptotes.
- Read off the x-values where the required graph is above or below.

Example 5

$$e^x < x + 2$$



Using GC, intersections at $x = -1.8414$, $x = 1.1462$

$$\therefore\ -1.84 < x < 1.15$$

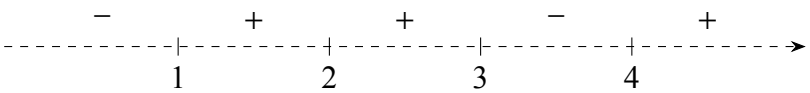
2.2 Sign Test Technique

We solve rational inequalities by finding where the expression changes sign.

- Make the right-hand side 0.
- Do **not** cross multiply unless the expression or bracket is always positive.
- Expressions that are always positive can be divided across, e.g. a quadratic with discriminant < 0 and positive leading coefficient: $x^2 + 2x + 5 = (x+1)^2 + 4 > 0$.

Example 6

$$\frac{(x-1)(x-2)^2}{(x-3)(x-4)} \leqslant 0$$



$x=2$: numerator $=0$ (included)

$x=3,\ x=4$: denominator $=0$ (excluded)

$\therefore x\leqslant 1$ or $x=2$ or $3<x<4$

> **Note**
>
> Crossing a factor with an odd power changes the sign; crossing a factor with an even power keeps the sign the same.

2.3 Common Substitutions

Once an inequality in x is solved, we can replace x with an expression to deduce the solution of a related inequality.

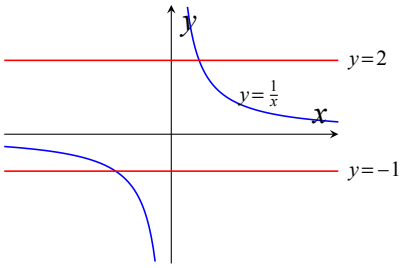
Replace x by	**Original inequality**		
	$x>2$	$x<-1$	$-1<x<2$
e^x	$x>\ln 2$	no solution	$x<\ln 2$
$\ln x$	$x>e^2$	$0<x<e^{-1}$	$e^{-1}<x<e^2$
$-x$	$x<-2$	$x>1$	$-2<x<1$
$x+1$	$x>1$	$x<-2$	$-2<x<1$
$\lvert x\rvert$	$x<-2$ or $x>2$	no solution	$-2<x<2$
x^2	$x<-\sqrt{2}$ or $x>\sqrt{2}$	no solution	$-\sqrt{2}<x<\sqrt{2}$
$\frac{1}{x}$	$\left(0<x<\frac{1}{2}\right)$	$(-1<x<0)$	$\left(x<-1 \text{ or } x>\frac{1}{2}\right)$
$\cos x,\ 0\leqslant x\leqslant 2\pi$	(no solution)	(no solution)	$(0\leqslant x<\pi$ or $\pi<x\leqslant 2\pi)$

Regardless of whatever substitution you use, using a graph will always work, but some can be solved without a graph.

For $\frac{1}{x}$, $\sin x$ and $\cos x$, draw the graph to solve.

Example 7

$$\frac{1}{x}>2, \quad \frac{1}{x}<-1, \quad -1<\frac{1}{x}<2$$



Example 8

$$\cos x>\tfrac{1}{2}, \quad \cos x<-\tfrac{1}{2},$$

$$-\tfrac{1}{2}<\cos x<\tfrac{1}{2}, \quad 0\leqslant x\leqslant 2\pi$$

